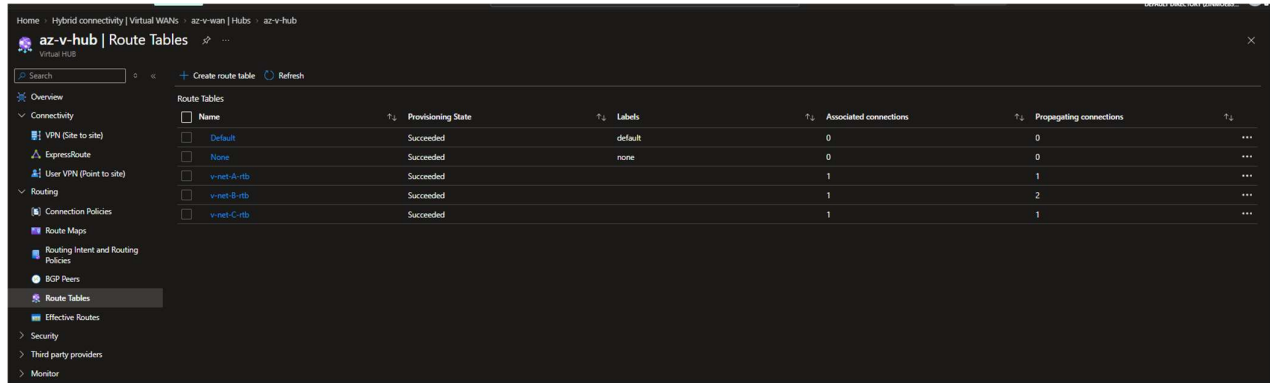


Azure Virtual WAN

Association & Propagation Lab

Step-by-step procedure using the actual Azure Portal lab screenshots



Name	Provisioning State	Labels	Associated connections	Propagating connections
Default	Succeeded	default	0	0
None	Succeeded	none	0	0
v-net-A-rtb	Succeeded		1	1
v-net-B-rtb	Succeeded		1	2
v-net-C-rtb	Succeeded		1	1

Lab topology: VNet A ↔ VNet B ↔ VNet C, with A and C isolated by route visibility

1. Lab Objective

This procedure demonstrates Azure Virtual WAN hub route-table association and propagation using three virtual networks. The objective is to allow VNet A to communicate with VNet B, and VNet B to communicate with VNet C, while VNet A and VNet C do not learn routes to each other.

Virtual Network	Address Space	Required Connectivity
v-net-A	10.1.0.0/16	Communicate with B
v-net-B	10.2.0.0/16	Communicate with A and C
v-net-C	10.3.0.0/16	Communicate with B

2. Association and Propagation - Clear Mental Model

Association = Which route table does this connection USE?

A VNet connection is associated with one hub route table. When traffic enters the Virtual Hub from that VNet connection, the associated route table is the routing view used to decide where the traffic can go.

```
VNet A connection --ASSOCIATE--> v-net-A-rtb
```

Propagation = Which route tables LEARN my routes?

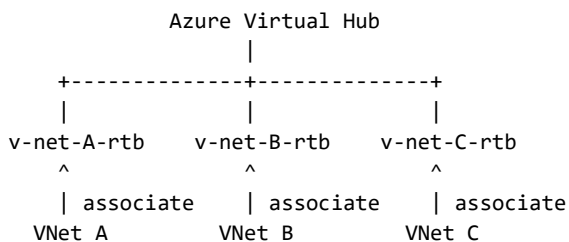
Propagation controls which hub route tables receive routes from a connection. If VNet B propagates to v-net-A-rtb, that table learns that 10.2.0.0/16 is reachable through the VNet B connection.

```
VNet B 10.2.0.0/16 --PROPAGATE--> v-net-A-rtb
```

Memory trick: Association = USE. Propagation = LEARN.

3. Final Routing Design

Connection	Associated Route Table	Propagates To
v-net-A-connection	v-net-A-rtb	v-net-B-rtb
v-net-B-connection	v-net-B-rtb	v-net-A-rtb, v-net-C-rtb
v-net-C-connection	v-net-C-rtb	v-net-B-rtb



Propagation:

```
A (10.1/16) ---> B route table
B (10.2/16) ---> A route table + C route table
C (10.3/16) ---> B route table
```

Result: A <----> B <----> C
A X----- C

4. Procedure

Step 1 - Create the Virtual WAN

Open Azure Portal, go to Hybrid connectivity > Virtual WANs, and create/open the Virtual WAN used for the lab. In this lab it is named az-v-wan.

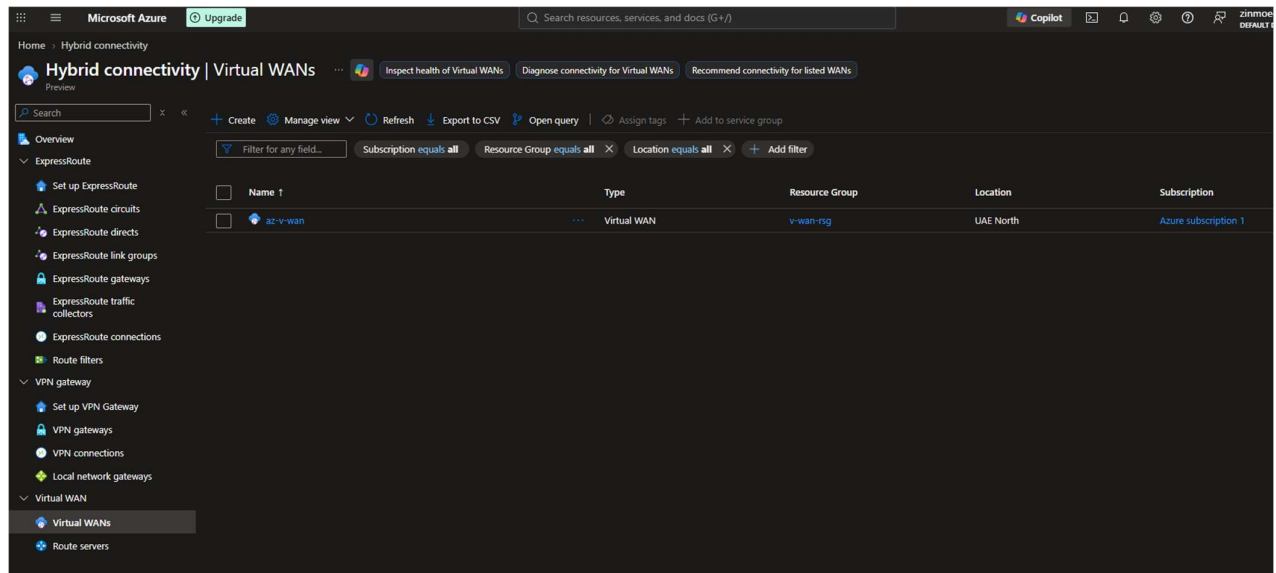


Figure 1 - Virtual WAN az-v-wan

Step 2 - Create the Virtual Hub

Open az-v-wan > Hubs > New Hub. The lab uses the following values:

- Region: UAE North
- Hub name: az-v-hub
- Hub private address space: 10.0.99.0/24
- Virtual hub capacity: 4 Routing Infrastructure Units

Microsoft Azure

Upgrade

Search resources, services, and docs

Home > Hybrid connectivity | Virtual WANs > az-v-wan | Hubs

Create virtual hub

Basics

Site to site

Point to site

ExpressRoute

Tags

Review + create

A virtual hub is a Microsoft-managed virtual network. The hub contains various service endpoints to enable connectivity from your on-premises network (vpnsite). [Learn more](#)

Project details

The hub will be created under the same subscription and resource group as the vWAN.

Subscription

Azure subscription 1

Resource group

v-wan-rsg

Virtual Hub Details

Region *

UAE North

Name *

az-v-hub

Hub private address space * ⓘ

10.0.99.0/24

Virtual hub capacity * ⓘ

4 Routing Infrastructure Units, 4 Gbps Router, Supports 4000 VMs

Hub routing preference * ⓘ

ExpressRoute

Creating a hub with a gateway will take 30 minutes.

Review + create

Previous

Next : Site to site >

Figure 2 - Virtual Hub creation

After deployment, confirm the hub status is Succeeded.

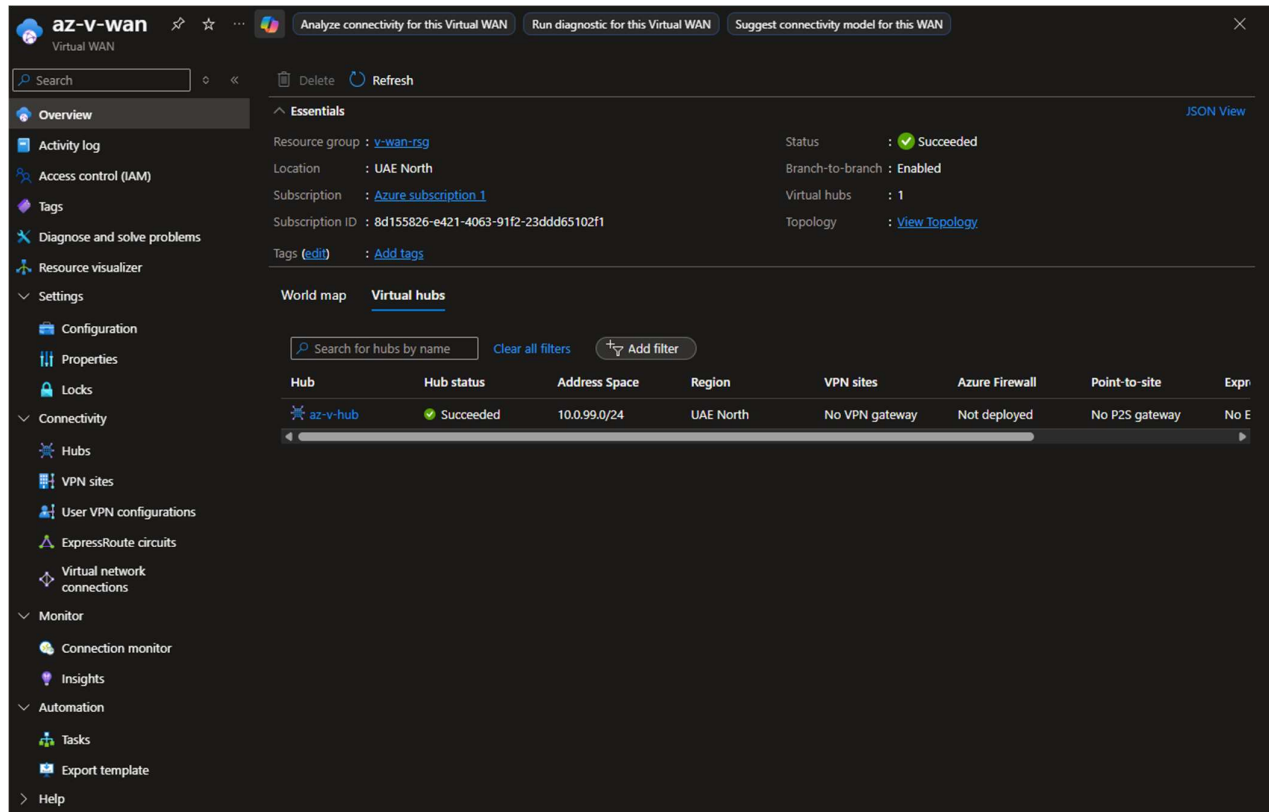


Figure 3 - az-v-hub successfully provisioned

Step 3 - Create three custom Hub Route Tables

Inside az-v-hub, open Routing > Route Tables. Create v-net-A-rtb, v-net-B-rtb, and v-net-C-rtb. For this lab, leave the manual route fields blank. The VNet prefixes will be learned through propagation.

Microsoft Azure

Upgrade

Search resources, services, and docs (G+)

[Home](#) > [Hybrid connectivity | Virtual WANs](#) > [az-v-wan](#) > [az-v-hub](#) | [Route Tables](#)

Create Route Table

Basics

Labels

Associations

Propagations

Project details

Subscription

Azure subscription 1

Resource group

v-wan-rsg

Instance details

Name *

v-net-A-rtb

⚠ Branch routes apply to all connected VPN sites, ExpressRoute circuits and User VPN connections. Destination prefix can be aggregated address or list of all branch prefixes

Route name	Destination type	Destination prefix	Next hop	Next Hop IP
	CIDR			

Review + create

Previous

Next : Labels >

Figure 4 - Creating v-net-A-rtb

Microsoft Azure

Upgrade

Search resources, services, and docs

Home > Hybrid connectivity | Virtual WANs > az-v-wan > az-v-hub | Route Tables

Create Route Table

Basics

Labels

Associations

Propagations

Project details

Subscription

Azure subscription 1

Resource group

v-wan-rsg

Instance details

Name

v-net-B-rtb

⚠ Branch routes apply to all connected VPN sites, ExpressRoute circuits and User VPN connections. Destination prefix can be aggregated address or list of all branch prefixes

Route name	Destination type	Destination prefix	Next hop	Next Hop IP
	CIDR			

Review + create

Previous

Next : Labels >

Figure 5 - Creating v-net-B-rtb

Microsoft Azure Upgrade

Home > Hybrid connectivity | Virtual WANs > az-v-wan > az-v-hub | Route Tables

Create Route Table

Basics Labels Associations Propagations

Project details

Subscription: Azure subscription 1

Resource group: v-wan-rsg

Instance details

Name: v-net-C-rtb

⚠ Branch routes apply to all connected VPN sites, ExpressRoute circuits and User VPN connections. Destination prefix can be aggregated address or list of all branch prefixes

Route name	Destination type	Destination prefix	Next hop	Next Hop IP
	CIDR			

Review + create Previous Next : Labels >

Figure 6 - Creating v-net-C-rtb

Confirm all route tables show Succeeded before creating the VNet connections.

Microsoft Azure Upgrade

Home > Hybrid connectivity | Virtual WANs > az-v-wan > az-v-hub

az-v-hub | Route Tables

Search Create route table Refresh

Overview

Connectivity

- VPN (Site to site)
- ExpressRoute
- User VPN (Point to site)

Routing

- Connection Policies
- Route Maps
- Routing intent and Routing Policies
- BGP Peers
- Route Tables**
- Effective Routes

Security

- Third party providers
- Monitor

Name	Provisioning State	Labels	Associated connections	Propagating connections
Default	Succeeded	default	0	0
None	Succeeded	none	0	0
v-net-A-rtb	Succeeded		0	0
v-net-B-rtb	Succeeded		0	0
v-net-C-rtb	Updating		0	0

Figure 7 - Custom route tables created

Step 4 - Connect VNet A

Go to az-v-wan > Virtual network connections > Add connection. Configure VNet A as follows:

Azure Virtual WAN Association & Propagation Lab Procedure

- Connection name: v-net-A-connection
- Virtual network: v-net-A
- Propagate to none: No
- Associate Route Table: v-net-A-rtb
- Propagate to Route Tables: v-net-B-rtb
- Propagate to labels: none
- Static routes: leave empty
- Bypass Next Hop IP: No

Figure 8 - VNet A connection routing configuration

Meaning: A USES v-net-A-rtb. A's 10.1.0.0/16 route is LEARNED by v-net-B-rtb.

Step 5 - If Azure reports the hub routing state is not Provisioned

If the portal displays the message below, wait for the Virtual Hub routing configuration and route-table updates to finish. Refresh the hub and Route Tables pages. Continue when the routing state is provisioned and all route tables show Succeeded. If the hub enters a Failed state, investigate/reset the router rather than repeatedly creating connections.

Figure 9 - Routing state not yet provisioned

Step 6 - Connect VNet B

VNet B is the middle network, so both A and C must learn B's route.

- Connection name: v-net-B-connection
- Virtual network: v-net-B
- Propagate to none: No
- Associate Route Table: v-net-B-rtb
- Propagate to Route Tables: select v-net-A-rtb AND v-net-C-rtb
- Do not select Default or v-net-B-rtb
- Static routes: leave empty
- Bypass Next Hop IP: No

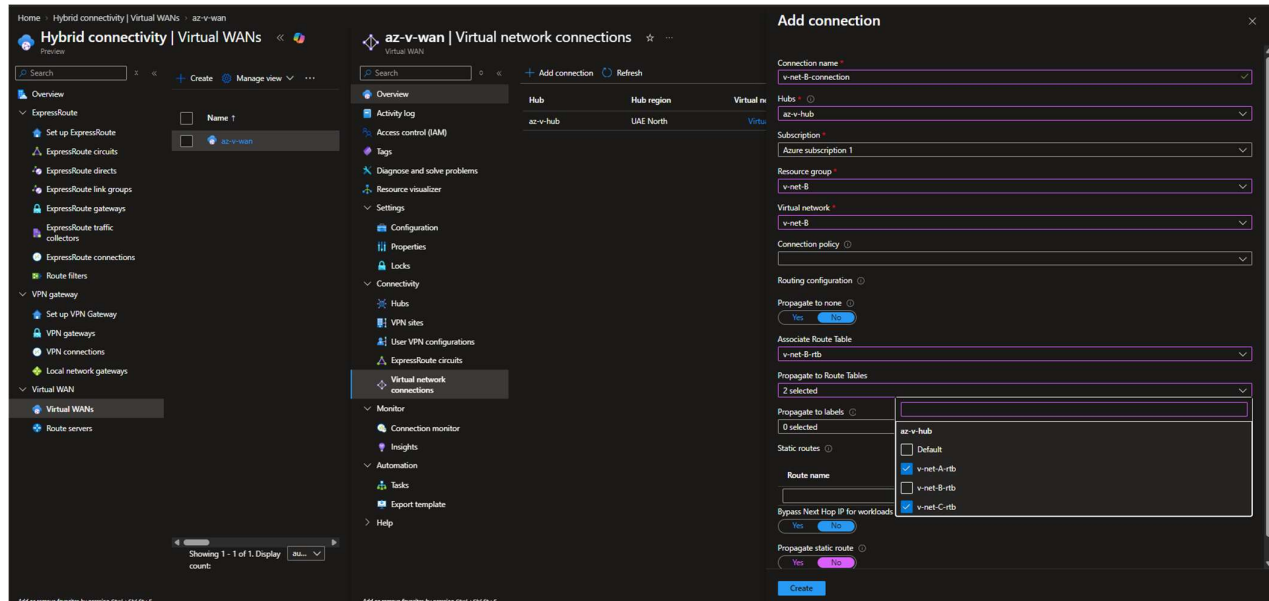


Figure 10 - VNet B propagates to both A and C route tables

Meaning: B USES v-net-B-rtb. Both A and C LEARN the B prefix 10.2.0.0/16.

Step 7 - Connect VNet C

- Connection name: v-net-C-connection
- Virtual network: v-net-C
- Propagate to none: No
- Associate Route Table: v-net-C-rtb
- Propagate to Route Tables: v-net-B-rtb only
- Propagate to labels: none
- Static routes: leave empty
- Bypass Next Hop IP: No

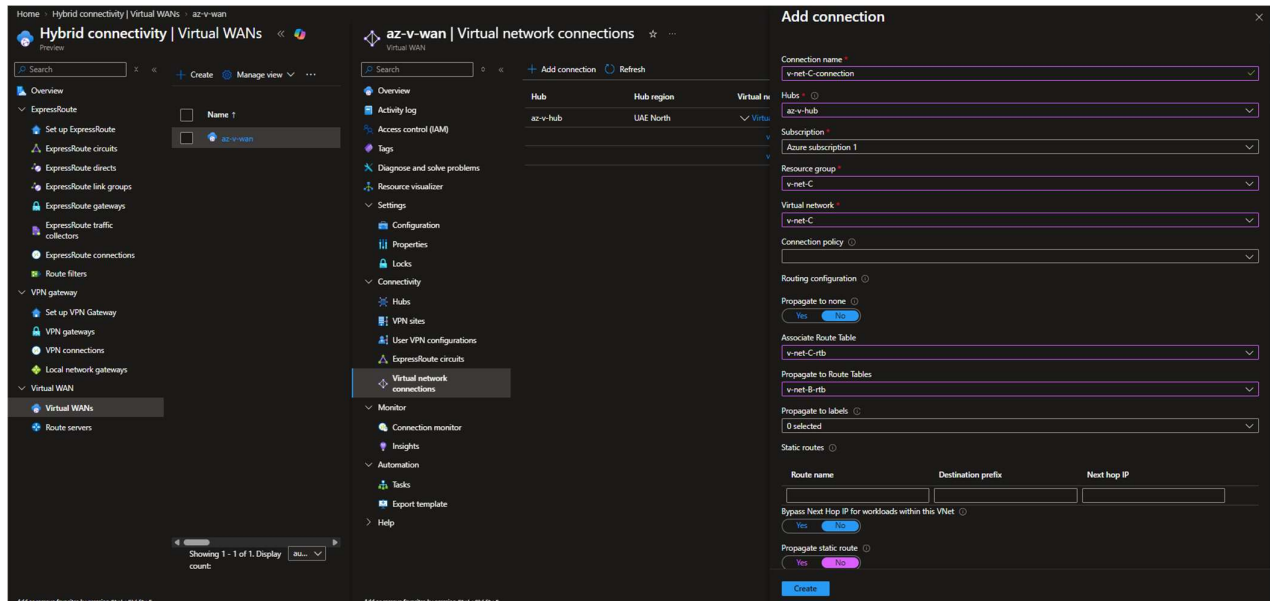


Figure 11 - VNet C connection routing configuration

Meaning: C USES v-net-C-rtb. C's 10.3.0.0/16 route is LEARNED by B, not A.

5. Verification

Open az-v-hub > Routing > Route Tables. The Associated connections and Propagating connections columns should match the following values.

Route Table	Associated	Propagating	Expected source(s)
v-net-A-rtb	1	1	B
v-net-B-rtb	1	2	A and C
v-net-C-rtb	1	1	B

Name	Provisioning State	Labels	Associated connections	Propagating connections
Default	Succeeded	default	0	0
None	Succeeded	none	0	0
v-net-A-rtb	Succeeded		1	1
v-net-B-rtb	Succeeded		1	2
v-net-C-rtb	Succeeded		1	1

Figure 12 - Final route-table association and propagation counts

6. Expected Effective Route View

After propagation converges, the important cross-VNet routes should conceptually be:

v-net-A-rtb
 10.2.0.0/16 -> VNet B
 No 10.3.0.0/16 route from C

```
v-net-B-rtb
10.1.0.0/16 -> VNet A
10.3.0.0/16 -> VNet C

v-net-C-rtb
10.2.0.0/16 -> VNet B
No 10.1.0.0/16 route from A
```

Use Routing > Effective Routes to verify the actual routes Azure has installed. Additional local/system routes can appear; focus on the cross-VNet prefixes above.

7. Packet-Flow Examples

A to B - Allowed by route visibility

VNet A -> v-net-A-rtb -> learned 10.2.0.0/16 route -> VNet B

B to C - Allowed by route visibility

VNet B -> v-net-B-rtb -> learned 10.3.0.0/16 route -> VNet C

A to C - No propagated C route in A's table

VNet A -> v-net-A-rtb -> no learned 10.3.0.0/16 route -> no intended A-to-C path

8. Important Security Note

Hub route-table isolation is a routing design, not a stateful firewall policy. If the requirement is a strict security rule that A and C must never communicate, enforce that requirement with appropriate NSGs, Azure Firewall, or another NVA/security control in addition to validating the routing design.

9. Final Memory Summary

ASSOCIATION = Which route table does my connection USE?

PROPAGATION = Which route tables LEARN my connection's routes?

For this lab: A associates to A's table and propagates to B's table; B associates to B's table and propagates to both A and C; C associates to C's table and propagates to B's table.